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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
VII Semester B. E. Examinations March-2022
Computer Science and Engineering
COMPUTER GRAPHICS AND VIRTUAL REALITY

*Time: 03 Hours**Maximum Marks: 100***Instructions to candidates:**

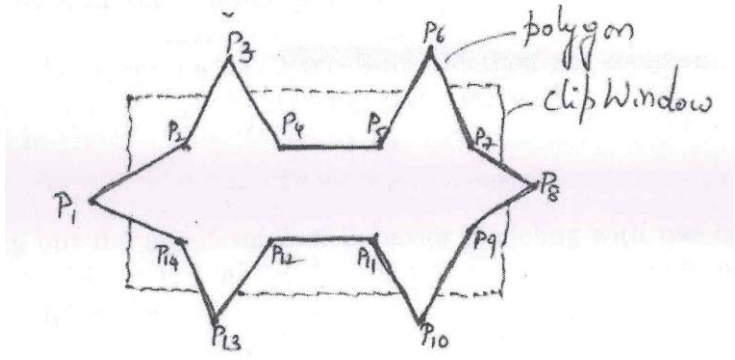
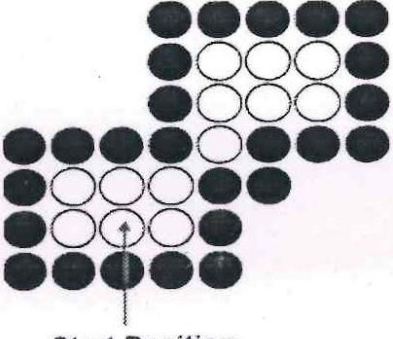
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	Write the steps in generation of 2D-Sierpinski gasket using the Iterative approach.	02
	1.2	Suppose on RGB raster system is designed using an 8 inch by 10 inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 6 bits per pixel in the frame buffer, how much storage (in bytes) is required for the frame buffer? Also determine the height of resized image 1023x768 to one that is 640 pixels wide with the same aspect ratio.	02
	1.3	A cube is drawn using two quadrilaterals(GL_QUADS) for its top and bottom face, and one quad strip(GL_QUAD_STRIP) for the front, left, back and right face. What is the minimum number of glVertex calls necessary to draw the cube this way?	02
	1.4	Differentiate between Flood fill and Boundary Fill algorithms for area filling.	02
	1.5	Show that transformation matrix for a reflection about a line $Y=X$ is equivalent to the reflection to X-axis by counter clockwise rotation of 90° .	02
	1.6	Differentiate between object space and image space methods of hidden surface removal.	02
	1.7	Give the complete taxonomy of classification of geometric projections.	02
	1.8	Name the various methods of modelling 3D object Geometry.	02
	1.9	_____ visibility-detection method is often referred to as the Painter's algorithm.	01
	1.10	Parametric representation of the line segment is given by $x = 2 + 4t, y = 4$ where $0 \leq t \leq 1$, Write its equivalent slope intercept form.	01
	1.11	List the five classic components of a Virtual Reality (VR) system.	01
	1.12	_____ is a technique performed in rasterizing stage of the graphics pipeline in order to modify the object model's surface properties such as color, specular reflection or pixel normals.	01

PART-B

2	a Discuss the major groups of graphics functions. Explain the same. b Write complete OpenGL 'C' program to recursively subdivide a tetrahedron to generate a 3D Sierpinski gasket. Note : Use Volume Subdivision only for generating 3D Sierpinski gasket. c Discuss the three I's of Virtual Reality.	05 07 04
3	a Discuss the various classes of logical input devices available to a developer of graphical applications. b Develop an OpenGL program to create the following hierarchical menu given in Fig. 3.b and attach the corresponding operations to each menu with mouse left button. You must give the appropriate code for menu entries. Fig. 3.b c What are Three-Dimensional Position Trackers. Differentiate between the various types of trackers(w.r.to working principle) used in Virtual Reality.	05 07 04
4	OR a Compare the different modes by which input devices provide input to an application program with schematic diagrams. b Write an OpenGL program to demonstrate the usage of glutMouseFunc() and glutKeyboardFunc() API's to draw Fig. 4.b using the primitive GL_LINE_LOOP whenever the mouse left button is pressed or whenever the character 'd' or 'D' is pressed from keyboard. Fig. 4.b c What are Sound displays? Discuss the significance of different categories of sound displays with a suitable real-world use cases for each type.	05 07 04
5	a Digitize a line from (1,2) to (12,18) on a raster screen using Bresenham's line drawing algorithm. Tabulate the results and represent it pictorially.	08

b	Examine the working principle of Sutherland-Hodgemen Polygon clipping algorithm for the polygon given in Fig. 5.b against the clipping window. Hand simulate all the intermediate processing steps and indicate the same with neat sketches.  Fig. 5.b OR	08
6 a	Examine the working principle of Boundary-Fill using a) 4-connected and b) 8-connected region (recursive approaches) for the polygon shown in Fig. 6.a. Illustrate the filling of the polygon using both the approaches pictorially. Indicate the complete trace of the same using appropriate data structure.  Fig. 6.a b Let R be the rectangle window whose lower left hand is at L(1,2) and upper right hand corner is at R(9,8). Apply Liang Barsky Line Clipping algorithm to clip the following line segments. Line 1 from A(11,6) to B(11,10) Line 2 from C(3,7) to D(3,10) Line 3 from E(,3) to F(8,4) Line 4 from I(-1,7) to J(11,1)	08
7 a	Illustrate with neat sketches how the region code approach of 2D clipping can be extended to three-dimensional point and line clipping. b Design and develop an interactive graphics program using appropriate OpenGL transformation functions to perform the following operations (independently) on a 3D-Cube: • Translate the cube with key press 'T' or 't' • Orient it with key press 'R' or 'r' • Zoom in and Zoom out with middle button(mouse) The output must be adjusted suitably when the window is resized or moved to a new position.	08

8	a	Bring out the significance of Behavior Modelling with use cases. Discuss about virtual human (or agent), group of such agents and behavior.	05
	b	Discuss the stages of Physical Modeling pipeline with appropriate mathematical formulations.	07
	c	Write the schematic diagram illustrating the various aspects of VR modelling cycle. Briefly describe the different components involved.	04